Document II: Background Questions

Participant identifier: Anonymous 123

1. Specify your gender (mark one)

☐ Female ☒ Male ☐ Not listed: _____ ☐ Prefer Not to Answer

2. Specify your level of education (mark one)

☐ Undergraduate Student

☐ Master Student

☒ PhD Student

☐ PhD

☐ Other _____

3. You currently work in...

☒ Academia

☐ Industry

☐ Research center

☐ Student

☐ Other _____

4. State your experience (in years) with software development 13

5. State your experience (in years) with modeling languages 13

6. I was familiar with the Blocky Maze game prior to this experiment: ☐ Yes ☒ No

Document III: Perception Questionnaire

Participant identifier: Anonymous 123

1. Questions about the experiment

	strongly agree → ← strongly disagree				
1. The modeling tasks were easy to complete	1	2	3	**4**	5
2. I had enough time to complete the tasks	1	2	3	4	**5**
3. I revised my models many times	1	2	**3**	4	5
4. I am satisfied with my final models	1	**2**	3	4	5
5. I lacked information or guidance while modeling	**1**	2	3	4	5

2. Usability of the environment

In the following questions, the word *"environment"* refers to the **modeling environment you used to complete the Block game.**

1. I think that I would like to use this environment frequently	**1**	2	3	4	5
2. I found the environment unnecessarily complex	**1**	2	3	4	5
3. I thought the environment was easy to use	1	2	3	4	**5**
4. I think that I would need the support of a technical person to be able to use this environment	**1**	2	3	4	5
5. I found the various functions in this environment were well integrated	1	2	3	4	**5**
6. I thought there was too much inconsistency in this environment	**1**	2	3	4	5
7. I would imagine that most people would learn to use this environment very quickly	1	2	3	4	**5**
8. I found the environment very cumbersome to use	**1**	2	3	4	5
9. I felt very confident using the environment	1	2	3	4	**5**
10. I needed to learn a lot of things before I could get going with this environment	**1**	2	3	4	5

3. Open questions

1. **How did you approach the modeling task? Describe the steps you followed.**

Look at the maze
Identify shortest path
Try to abbreviate path logic
Run
Fix errors
Improve

2. **Which part of the task was most difficult, and why?**

Maze 10 due to complexity of the maze.
Finding improvements had several possible solutions

3. **What kind of support would have helped you most during the task?**

Design Space exploration of alternate solutions path, i.e., instead of following all right turns first follow all left turns.
Nothing with regards of environment or tool.

Document II: Background Questions

Participant identifier: Dennis

1. Specify your gender (mark one)

☐ Female ☑ Male ☐ Not listed: _____ ☐ Prefer Not to Answer

2. Specify your level of education (mark one)

☐ Undergraduate Student

☑ Master Student

☐ PhD Student

☐ PhD

☐ Other _____

3. You currently work in...

☑ Academia

☐ Industry

☐ Research center

☐ Student

☐ Other _____

4. State your experience (in years) with software development 4 years

5. State your experience (in years) with modeling languages 0 years

6. I was familiar with the Blocky Maze game prior to this experiment: ☑ Yes ☐ No

Document III: Perception Questionnaire

Participant identifier: Dennis

1. Questions about the experiment

	strongly agree → ← strongly disagree				
1. The modeling tasks were easy to complete	1	2	3	4	5
2. I had enough time to complete the tasks	1	2	3	4	5
3. I revised my models many times	1	2	3	4	5
4. I am satisfied with my final models	1	2	3	4	5
5. I lacked information or guidance while modeling	1	2	3	4	5

2. Usability of the environment

In the following questions, the word *"environment"* refers to the **modeling environment you used to complete the Block game.**

1. I think that I would like to use this environment frequently	1	2	3	4	5
2. I found the environment unnecessarily complex	1	2	3	4	5
3. I thought the environment was easy to use	1	2	3	4	5
4. I think that I would need the support of a technical person to be able to use this environment	1	2	3	4	5
5. I found the various functions in this environment were well integrated	1	2	3	4	5
6. I thought there was too much inconsistency in this environment	1	2	3	4	5
7. I would imagine that most people would learn to use this environment very quickly	1	2	3	4	5
8. I found the environment very cumbersome to use	1	2	3	4	5
9. I felt very confident using the environment	1	2	3	4	5
10. I needed to learn a lot of things before I could get going with this environment	1	2	3	4	5

3. Open questions

1. **How did you approach the modeling task? Describe the steps you followed.**

I tried to sketch the plain simple diagram / instruction & then add loops to it. Sketch out simple pseudo code.

2. **Which part of the task was most difficult, and why?**

I found time limit as constraint to ~~the problem~~ for solving all the problems.

3. **What kind of support would have helped you most during the task?**

maybe a wall clock running along with the score board would have helped to motivate the participant.

Document II: Background Questions

Participant identifier: Ghost

1. **Specify your gender (mark one)**

   ☐ Female ☒ Male ☐ Not listed: _____ ☐ Prefer Not to Answer

2. **Specify your level of education (mark one)**

   ☐ Undergraduate Student

   ☒ Master Student

   ☐ PhD Student

   ☐ PhD

   ☐ Other _____

3. **You currently work in...**

   ☒ Academia

   ☐ Industry

   ☐ Research center

   ☐ Student

   ☐ Other _____

4. **State your experience (in years) with software development** 0

5. **State your experience (in years) with modeling languages** 1

6. **I was familiar with the Blocky Maze game prior to this experiment:** ☒ Yes ☐ No

Document III: Perception Questionnaire

Participant identifier: Ghost

1. Questions about the experiment

	strongly agree → ← strongly disagree				
1. The modeling tasks were easy to complete	1	**2**	3	4	5
2. I had enough time to complete the tasks	1	2	3	**4**	5
3. I revised my models many times	1	2	3	4	**5**
4. I am satisfied with my final models	1	2	3	**4**	5
5. I lacked information or guidance while modeling	**1**	2	3	4	5

2. Usability of the environment

In the following questions, the word *"environment"* refers to the **modeling environment you used to complete the Block game.**

1. I think that I would like to use this environment frequently	1	2	**3**	4	5
2. I found the environment unnecessarily complex	1	**2**	3	4	5
3. I thought the environment was easy to use	1	2	3	**4**	5
4. I think that I would need the support of a technical person to be able to use this environment	**1**	2	3	4	5
5. I found the various functions in this environment were well integrated	1	2	3	**4**	5
6. I thought there was too much inconsistency in this environment	1	**2**	3	4	5
7. I would imagine that most people would learn to use this environment very quickly	1	2	3	4	**5**
8. I found the environment very cumbersome to use	**1**	2	3	4	5
9. I felt very confident using the environment	1	2	3	**4**	5
10. I needed to learn a lot of things before I could get going with this environment	**1**	2	3	4	5

3. Open questions

1. **How did you approach the modeling task? Describe the steps you followed.**

First, I looked at the maze and searched fw repeting patterns. If pattern existed to some extent and in sequence, put the respective actions in if-loop. If not or after the pattern, diffrient patten followed, assigned respective actions fw them. fw difficult routes, strategy had to be improvised accordingly fw blocks.

2. **Which part of the task was most difficult, and why?**

In some levels of the maze, some alternate routes lead to dead end. That I believe was the challenge to tackle in ~~available~~ optimum number of operations.

3. **What kind of support would have helped you most during the task?**

NA. All neecssary support was provided.

Document II: Background Questions

Participant identifier: player-697

1. Specify your gender (mark one)

☐ Female ☒ Male ☐ Not listed: _____ ☐ Prefer Not to Answer

2. Specify your level of education (mark one)

☐ Undergraduate Student

☒ Master Student

☐ PhD Student

☐ PhD

☐ Other _____

3. You currently work in...

☒ Academia

☐ Industry

☐ Research center

☐ Student

☐ Other _____

4. State your experience (in years) with software development 10

5. State your experience (in years) with modeling languages 5

6. I was familiar with the Blocky Maze game prior to this experiment: ☐ Yes ☒ No

Document III: Perception Questionnaire

Participant identifier: player_697

1. Questions about the experiment

	strongly agree → ← strongly disagree				
1. The modeling tasks were easy to complete	1	2	3	4	5
2. I had enough time to complete the tasks	1	2	3	4	5
3. I revised my models many times	1	2	3	4	5
4. I am satisfied with my final models	1	2	3	4	5
5. I lacked information or guidance while modeling	1	2	3	4	5

2. Usability of the environment

In the following questions, the word *"environment"* refers to the **modeling environment you used to complete the Block game.**

1. I think that I would like to use this environment frequently	1	2	3	4	5
2. I found the environment unnecessarily complex	1	2	3	4	5
3. I thought the environment was easy to use	1	2	3	4	5
4. I think that I would need the support of a technical person to be able to use this environment	1	2	3	4	5
5. I found the various functions in this environment were well integrated	1	2	3	4	5
6. I thought there was too much inconsistency in this environment	1	2	3	4	5
7. I would imagine that most people would learn to use this environment very quickly	1	2	3	4	5
8. I found the environment very cumbersome to use	1	2	3	4	5
9. I felt very confident using the environment	1	2	3	4	5
10. I needed to learn a lot of things before I could get going with this environment	1	2	3	4	5

3. Open questions

1. **How did you approach the modeling task? Describe the steps you followed.**

I tried to find patterns in the movement (path) & then tried to replicate it using loops.
If I didn't find a pattern I ~~e~~ built the path by hand.
The last level I found the pattern but didn't manage to build it nicely.

2. **Which part of the task was most difficult, and why?**

Interface is very easy & straight forward.
Since I already know how to program, I was lacking "advanced" features like ~~a~~ negation.
Hence, using only the given blocks was most challenging for me.

3. **What kind of support would have helped you most during the task?**

Was perfect as is. Easy instructions & especially the showcase helped

Document II: Background Questions

Participant identifier: 784

1. **Specify your gender (mark one)**

   ☐ Female ☒ Male ☐ Not listed: _____ ☐ Prefer Not to Answer

2. **Specify your level of education (mark one)**

   ☐ Undergraduate Student

   ☐ Master Student

   ☐ PhD Student

   ☒ PhD

   ☐ Other _____

3. **You currently work in...**

   ☒ Academia

   ☐ Industry

   ☐ Research center

   ☐ Student

   ☐ Other _____

4. **State your experience (in years) with software development** 30

5. **State your experience (in years) with modeling languages** 0

6. **I was familiar with the Blocky Maze game prior to this experiment:** ☐ Yes ☒ No

Document III: Perception Questionnaire

Participant identifier: 784

1. Questions about the experiment

	strongly agree → ← strongly disagree				
1. The modeling tasks were easy to complete	1	2	3	4	5
2. I had enough time to complete the tasks	1	2	3	4	5
3. I revised my models many times	1	2	3	4	5
4. I am satisfied with my final models	1	2	3	4	5
5. I lacked information or guidance while modeling	1	2	3	4	5

2. Usability of the environment

In the following questions, the word *"environment"* refers to the **modeling environment you used to complete the Block game.**

1. I think that I would like to use this environment frequently	1	2	3	4	5
2. I found the environment unnecessarily complex	1	2	3	4	5
3. I thought the environment was easy to use	1	2	3	4	5
4. I think that I would need the support of a technical person to be able to use this environment	1	2	3	4	5
5. I found the various functions in this environment were well integrated	1	2	3	4	5
6. I thought there was too much inconsistency in this environment	1	2	3	4	5
7. I would imagine that most people would learn to use this environment very quickly	1	2	3	4	5
8. I found the environment very cumbersome to use	1	2	3	4	5
9. I felt very confident using the environment	1	2	3	4	5
10. I needed to learn a lot of things before I could get going with this environment	1	2	3	4	5

3. Open questions

1. **How did you approach the modeling task? Describe the steps you followed.**

Applied algorithmic thinking into these simple well known blocks

2. **Which part of the task was most difficult, and why?**

Number 10 – Difficult because of a lack of a better strategy, not because of usage of Blockly

3. **What kind of support would have helped you most during the task?**

None